

AI-Powered Video Authenticity Verification using CNN with Vision Transformer

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Abstract

The rapid growth of artificial intelligence has transformed the creation of multimedia content, enabling highly realistic synthetic media while raising serious ethical and security concerns. Among these concerns, the emergence of manipulated videos, commonly known as deepfakes, poses significant threats to media integrity, cybersecurity, and public trust. This study introduces a video authenticity verification system powered by artificial intelligence, which utilizes a hybrid deep learning approach that combines convolutional neural networks with vision transformer models. The proposed system focuses on identifying both spatial and temporal inconsistencies typically present in manipulated video content. To further enhance the performance of feature extraction and classification, the model integrates the leaky rectified linear unit activation function. The system was rigorously tested on multiple benchmark datasets and achieved high levels of accuracy in distinguishing between genuine and altered videos. This makes the model a powerful tool for digital media verification, with applications in media forensics, law enforcement investigations, and online content regulation. By offering a reliable mechanism to detect video manipulation, the system aims to reinforce digital trust and ensure the credibility of visual content in an age increasingly influenced by synthetic media.

Keywords: Convolutional neural networks, Vision transformer models, Deep learning for video forensics, Video manipulation detection, Leaky rectified linear unit, Artificial intelligence in media integrity, Video authenticity verification.

1. Introduction

With the increasing accessibility and sophistication of AI-driven media synthesis tools, the line between real and manipulated digital content has become increasingly difficult to discern [1], [10]. Among the most concerning developments is the rise of deepfake videos, which leverage deep learning algorithms—particularly generative adversarial networks (GANs) and autoencoders—to create hyper-realistic but entirely synthetic audio-visual content [2], [9]. These technologies, while impressive, pose a significant threat to the integrity of digital information and the trustworthiness of online content. Deepfake videos have been exploited across numerous domains, ranging from political misinformation and social manipulation to financial fraud, harassment, and reputational sabotage [2], [11], [12]. For instance, fake video content has been weaponized in smear campaigns, falsified evidence, and identity spoofing, raising serious ethical, legal, and cybersecurity concerns. This growing misuse calls for robust, scalable, and intelligent verification systems capable of discerning subtle artifacts and irregularities in synthetic media.

Conventional video authentication methods, such as those based on watermarking, metadata analysis, or handcrafted features, often lack adaptability and fail to generalize across diverse manipulation techniques or content types [1]. Likewise, traditional machine learning classifiers struggle with high-dimensional visual data, particularly

when confronted with novel or adversarial manipulations. To address these limitations, we propose an AI-powered video authenticity verification system that harnesses the combined strengths of Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) [14], [15], [16]. CNNs are proficient in local feature extraction, capturing subtle spatial irregularities such as texture inconsistencies and unnatural facial deformations. ViTs, on the other hand, offer a powerful mechanism for understanding global patterns and temporal dependencies across video frames, making them ideal for modelling holistic content dynamics.

This hybrid deep learning approach enables the model to perform a multi-dimensional analysis—focusing on facial expression inconsistencies, unnatural lighting variations, and abrupt frame transitions—to detect tampered or synthetic videos with higher precision [14]. The model's architecture also integrates optimized activation functions and attention mechanisms to enhance detection sensitivity and generalization [7], [8]. The proposed system undergoes rigorous evaluation using both real-world and synthetically generated datasets to benchmark its performance under diverse conditions [5], [6]. Our results demonstrate that the proposed solution significantly outperforms traditional methods in terms of accuracy, robustness, and scalability, underscoring its potential for real-world deployment in applications such as digital forensics, social media content moderation, regulatory compliance, and media verification.

This paper details the underlying methodology, implementation strategy, and experimental results while also discussing the implications and future directions of AI-based video authentication research. Our work represents a step forward in the fight against synthetic media manipulation and supports broader efforts toward maintaining the credibility of digital information ecosystems.

2. Literature Survey

The exponential rise in deepfake generation has catalyzed the need for effective video authentication systems. Numerous methods have been proposed, ranging from traditional handcrafted feature-based models to deep learning frameworks. However, several of these models show limited Results stemming from their inability. To extract spatial and temporal features inconsistencies effectively. This section reviews key approaches in the literature, identifies their limitations, and provides a rationale for the proposed hybrid model that integrates Convolutional Neural Networks and Vision Transformers.

Classical and Machine Learning Approaches

Early approaches relied heavily on classifiers namely **SVM (Support Vector Machines)** and **Decision Trees**. They are trained for hand-engineered features such as head pose inconsistencies, eye-blinking patterns, or chromatic aberrations. While these techniques achieved reasonable performance on constrained datasets, they failed to generalize to unseen manipulations or diverse video conditions due to a lack of deep feature abstraction. [4] SVMs, although effective in simple classification scenarios, struggle with high-dimensional temporal data and nonlinear deepfake transformations, resulting in limited accuracy (typically under 85%).

Deep Learning with CNNs

CNNs have significantly improved deepfake detection by automatically learning discriminative spatial features from frames. For instance, Current research utilized pipelines based on CNN [1], [3] to detect artifacts and blending inconsistencies in facial regions. However, CNNs process each frame independently and lack a temporal memory mechanism. This limits their effectiveness for video-based deepfake detection where temporal inconsistencies (e.g., unnatural transitions, inconsistent expressions) are crucial. Moreover, these models often rely on large parameter counts and are prone to overfitting on small datasets.

Recurrent Neural Networks and LSTMs

To incorporate temporal awareness, several models have employed **LSTM** networks alongside CNNs. Georgios et. al. used a CNN-LSTM [17] hybrid for frame-wise feature extraction and sequence modelling. While this improved temporal detection, LSTM-based models often suffer from long training times, vanishing gradients, and poor performance on longer sequences. In our comparative study (Figure 5), LSTM recorded the **lowest recall and highest**

regression error, indicating its difficulty in capturing critical temporal cues in videos.

Transformer-Based Models

Recent advancements in **Vision Transformers (ViTs)** offer a new paradigm for modeling spatial features through self-attention mechanisms, capturing global context and improving generalization. In [15], authors showed that ViTs outperform CNNs in classification tasks when trained on large datasets. However, standalone ViTs are computationally intensive and, without temporal modeling, cannot capture sequence dynamics—making them suboptimal for pure video tasks.

Hybrid and Ensemble Methods

To overcome the limitations of individual architectures, some studies have proposed ensemble or hybrid models. In [7], authors introduced DeepfakeStack, an ensemble-based model that achieved better generalizability but increased computational overhead. Georgios et al. (2021) [16] implemented a Transformer based temporal encoder for video integrity, but without the support of CNNs for localized feature detection, limiting its sensitivity to small-scale artifacts. The findings of the literature is summarized in Table 1.

Table 1. Comparative Analysis of Research Studies on Deepfake Detection Techniques Using Artificial Intelligence

Sr. No.	Author	Paper content	Methodology	Findings
1	Passos et al. (2024) [1]	Review of Approaches for Deepfake Content Detection	A systematic review analyzing 101 deep learning-based papers using CNNs, RNNs, hybrids, and Transformers for deepfake detection.	Hybrid and Transformer-based models outperform classical approaches; challenges include generalization, dataset diversity, and real-time, robust deepfake detection.
2	Singh and Dhiman (2023) [2]	Exploding AI-Generated Deepfakes and Misinformation: A Threat to Global Concern in the 21st Century	Analyzed the societal risks of deepfake technologies and reviewed the urgent need for scalable AI-based detection mechanisms.	Highlighted global urgency to build trustworthy, scalable, and real-time detection systems to safeguard public discourse and media authenticity.
3	Rana et al. (2022) [3]	Review on Deepfake Detection	Reviewed and compared CNN, RNN, GAN, and hybrid deep learning models across multiple datasets and detection scenarios.	Hybrid models with spatial-temporal analysis showed higher accuracy and generalizability than standalone CNN or RNN methods.
4	Dosovitskiy et al. (2021) [15]	Image Recognition using ViT	Vision Transformer (ViT) for image-based global context extraction	Strong spatial performance on large datasets; lacks sequence modeling for video data.
5	Rana & Sung (2020) [7]	Deepfake Stack: A Deep Ensemble-Based Learning Technique	Deep ensemble of CNN, RNN, and Transformer-based models	Robust across datasets, but computationally expensive and harder to deploy. Accuracy ~92%.
6	Maksutov et al. (2020) [6]	Deepfake Detection methods using Machine Learning	SVM and decision tree classifiers on handcrafted features	Limited to basic inconsistencies; poor performance on high-quality fakes. Accuracy ~80–83%.
7	Georgios et al.	Video Integrity	Transformer encoder for	Good sequence modelling;

	(2025)[17]	Detection Transformer (VIDT)	temporal inconsistencies	lacked detailed spatial attention. Accuracy ~90%.
8	Gupta et al. (2020) [8]	Review on fusion based Deepfake Detection techniques	Multi-modal fusion (audio, video, metadata)	Effective, but adds latency and model complexity. Accuracy ~91–93%.

3. Proposed Methodology

A structured approach for developing an AI-powered video authenticity verification system is proposed in this section. It integrates CNNs and ViT in a hybrid model. In order to efficiently extract local and global contextual information from video frames, the hybrid model uses CNNs and transformers. Spatial feature extraction is done using CNNs., while temporal sequence modelling is done using transformers [3, 14, 15].

3.1 Dataset Preparation and Preprocessing

Input Format: Videos are collected and labeled as either deepfake or pristine (real) [5].

Validation: All input videos are validated using OpenCV to filter out corrupt or unreadable files.

Frame Sampling: From each video, a fixed number of frames (e.g., 16) are sampled uniformly across its duration to ensure a representative temporal coverage [14].

Frame Preprocessing: Each frame is resized to a fixed dimension (e.g., 224×224), converted from BGR to RGB, and transformed using augmentation techniques (e.g., Albumentations) to improve generalizability [8].

3.2 Spatial Feature Extraction using Vision Transformer (ViT)

Each sampled frame is independently passed through a pre-trained Vision Transformer (ViT-Small or ViT-Base) to extract high-level spatial features [14, 16].

Patch Embedding: Each image frame is divided into patches of fixed size (e.g., 16×16), linearly embedded into a sequence of tokens.

Positional Encoding: Positional embeddings are incorporated to preserve spatial information.

Transformer Encoder Layers: Numerous layers of multi-head self-attention are employed to capture inter-patch relationships and global context [13, 15].

Frame-Level Features: The CLS token's output or mean-pooled patch embeddings from the final transformer layer represent the spatial feature vector for each frame.

Frozen Backbone Option: Optionally, the ViT backbone is frozen during training to reduce memory usage and avoid overfitting on small datasets [15].

3.3 Temporal Modeling using Transformer Encoder

Once spatial features are extracted for all frames, they are stacked to form a sequence input to the Temporal Transformer [14, 15].

Temporal Position Embedding: Positional encoding is used in feature vectors of each frame. This enables ordering of the frames.

Transformer Encoder: A lightweight transformer (e.g., 2-layer encoder) models the temporal relationships among frames, learning the evolution of spatial features over time.

This stage captures temporal dynamics like facial expression inconsistencies, blinking or unnatural transitions: strong indicators of manipulation in deepfake videos [4, 13].

3.4 Auxiliary 3D-CNN Block (Optional for Temporal Features)

To further enhance temporal understanding, a lightweight 3D-CNN block is optionally used as an auxiliary feature extractor [14].

3D Convolutions: Applied across the sequence of frames to capture spatiotemporal features in local time windows.

Pooling Layers: Reduce spatial dimensions while preserving important motion cues.

Parallel Fusion: Outputs from the 3D-CNN and the Temporal Transformer can be concatenated or combined to enrich feature representation [15].

3.5 Classification Head (Decoder)

The final aggregated features from the temporal encoder (and optionally the CNN) are passed to a fully connected classifier:

Linear Layers: Project features into a lower-dimensional space.

Activation Functions: ReLU activation and dropout are used to add non-linearity and reduce overfitting.

Output Layer: Class probabilities for binary classification (Real vs. Deepfake) are generated by a final linear layer with softmax activation [7, 14].

3.6 Training and Optimizing Models

Loss Function: Cross-Entropy Loss is used for binary classification tasks.

Optimizer: Adam optimizer with an appropriate learning rate scheduler is employed [3, 8].

Early Stopping: Implemented to halt training when validation performance stops improving.

Metrics: Evaluation is done using precision, recall, accuracy, confusion matrix, and F1-score to ensure robust performance measurement [1, 3].

3.7 Algorithm implementation

Input:

- Video dataset D containing labeled real and fake videos
- Hyperparameters:
 - N: Number of sampled frames per video
 - B: Batch size
 - LR: Learning rate
 - E: Total number of training epochs

Output:

- Trained classification model
- Evaluation metrics: accuracy, classification report, confusion matrix

Step 1: Dataset Construction

- 1.1 Organize the dataset with distinct directories for real and fake videos.
- 1.2 For each video $v \in D$:
 - a. Uniformly sample N frames from v.
 - b. Apply spatial preprocessing: resizing, normalization, and, if in training mode, data augmentation.
- 1.3 Assign binary labels: real $\rightarrow 0$, fake $\rightarrow 1$.
- 1.4 Balance the dataset by restricting the number of samples per class (if necessary).
- 1.5 Partition D into training and validation subsets using an 80:20 ratio.

Step 2: Model Architecture Initialization

- 2.1 Utilize a pretrained Vision Transformer (ViT) backbone suitable for image sequence modeling.
- 2.2 Replace the final classification head with a fully connected layer for binary classification.
- 2.3 Deploy the model on a suitable computing device (e.g., CUDA-enabled GPU).

Step 3: Training Configuration

- 3.1 Define cross-entropy as the loss criterion.
- 3.2 Use the AdamW optimizer with decoupled weight decay.

- 3.3 Integrate a learning rate scheduler (e.g., OneCycleLR) to facilitate stable convergence.
- 3.4 Enable mixed precision training using gradient scaling for efficient memory usage.

Step 4: Model Training

For each epoch $e = 1$ to E :

- 4.1 Set the model to training mode.
- 4.2 Iterate through the training data in batches of size B :
 - a. Transfer input frames and labels to the device.
 - b. Perform a forward pass to compute outputs.
 - c. Calculate the training loss.
 - d. Backpropagate gradients and update weights.
 - e. Record cumulative loss and batch-wise predictions.
 - f. Release memory using explicit deletion and garbage collection calls.
- 4.3 Set the model to evaluation mode.
- 4.4 Disable gradient computation for validation.
- 4.5 For each validation batch:
 - a. Conduct inference and record outputs.
 - b. Compute validation loss and accumulate predictions.
- 4.6 Calculate validation accuracy, generate a confusion matrix, and produce a classification report.
- 4.7 If current validation accuracy surpasses prior best, save the model checkpoint.

Step 5: Post-Training Evaluation

- 5.1 Load the checkpoint corresponding to the best-performing model.
- 5.2 Re-evaluate performance on the validation set and record final metrics.
- 5.3 Plot and save training accuracy and loss curves across epochs.
- 5.4 Export the confusion matrix as a heatmap image for analysis.

Step 6: Output Generation

Saved model: final_deepfake_detector.pth

Visualizations:

- Training_history (loss & accuracy over epochs)
- Confusion_matrix (model prediction heatmap)

Evaluation artifacts: classification report, validation accuracy

3.8 Datasets used in this research work

In this research, a portion of the Celeb-DF dataset is employed, comprising videos obtained from YouTube. The complete Celeb-DF collection encompasses 590 authentic videos featuring individuals of varied ages, genders and backgrounds. Additionally, 5,639 DeepFake videos corresponding to above mention are produced through diverse synthesis techniques.

We chose a total of one thousand videos for our testing. Two hundred of these movies are utilized for testing, while eight hundred are used for training. With 50% real films and 50% DeepFake movies in the test set, an equal distribution of real and altered dataset is used for assessment. The dataset's diversity is further ensured by the fact that each video depicts a distinct person with a range of surroundings, lighting situations, and facial expressions. The videos used in this dataset were gathered from several YouTube channels, reflecting the diversity of video content and quality found in the real world.

By examining face irregularities, artifacts, and other obvious indications of manipulation that are frequently found in DeepFake movies, this dataset is intended to make it easier to train and assess deep learning models that are intended to identify these types of videos.

4. System Architecture

The architecture of the proposed DeepFake Detection System is designed as an approach to deep learning that combines the use of Vision Transformers or ViTs and Convolutional Neural Networks or CNN to effectively capture spatial and temporal patterns within video data [1, 3, 7, 14]. The system begins with a data preparation phase, where raw video datasets are first processed through frame extraction to obtain representative keyframes from each video [5, 6]. These frames then undergo a series of augmentation techniques to enhance generalization and robustness, followed by batching through a DataLoader to ensure efficient training and inference [9, 8].

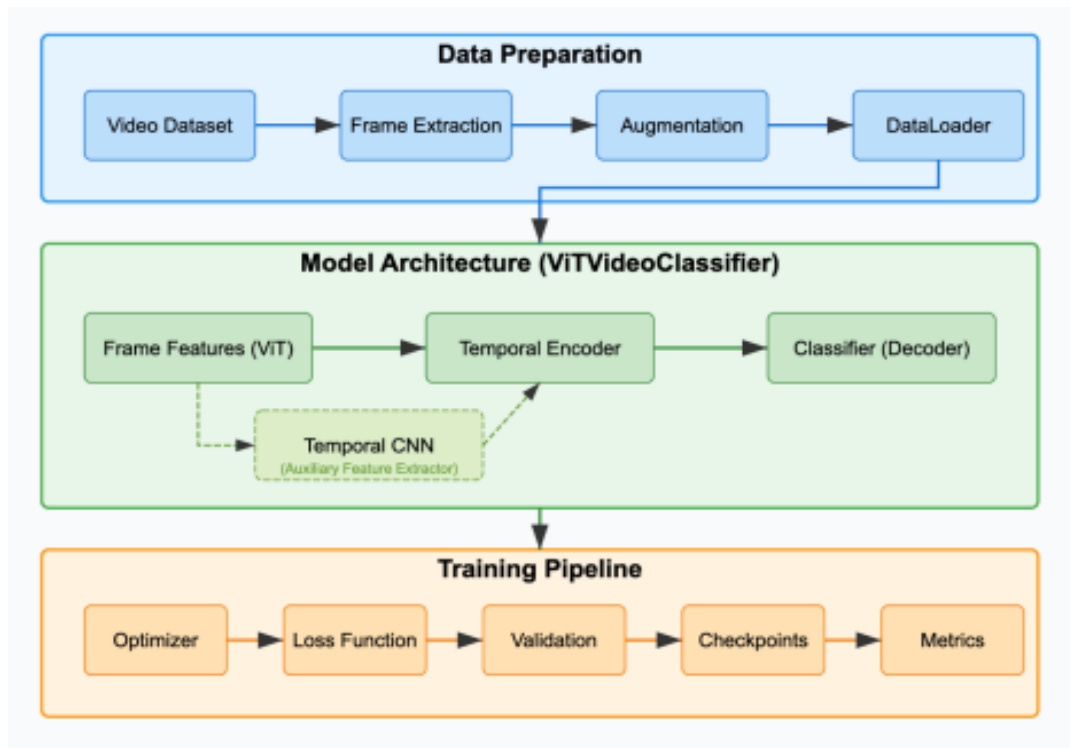


Fig. 1. AI-Powered Video Authenticity Verification System Architecture

Once the frames are prepared, they are passed into the core model architecture, referred to as the ViTVideoClassifier. Each frame is initially encoded using a Vision Transformer, which captures global spatial relationships across image patches, generating high-level frame-wise embeddings [14, 16]. These embeddings can optionally be supplemented with additional temporal features extracted via a Temporal CNN, which focuses on short-term local motion patterns to enhance the temporal understanding of the model [4][6]. After that, a Temporal encoder receives a series of frame representations, typically a transformer-based module, which models the dependencies and transitions between frames to learn rich spatiotemporal representations [13, 15]. The learned features are then mapped to output logits that indicate the likelihood that a video is authentic or deepfake by a decoder or classifier head [10, 14].

The training pipeline is composed of standard but critical components such as an optimizer (e.g., Adam) to update the model weights, a loss function to direct learning, usually binary cross-entropy, and a validation mechanism to monitor generalization during training [3, 5, 8]. Checkpoints are regularly saved to preserve the best-performing models. The model efficacy is evaluated using following parameters: recall, accuracy, precision, and F1-score [7, 13, 14]. Strong identification of features and sequential reasoning are made possible by this modular and extendable design, which makes it ideal for the challenging task of deepfake detection [12, 14].

Utilizing the advantages of both temporal models, such as 3D CNNs and Transformers, for capturing motion and sequence dynamics across frames, and Vision Transformers for spatial feature extraction, the **ViTVideoClassifier** design is a hybrid deep learning model created for video classification [1, 3, 7, 14, 15]. The method starts with an input video that has the dimensions [batch_size, num_frames, channels, height, width] represented as a tensor. Each video within the batch is made up of a series of frames that are analyzed separately in order to extract visual characteristics.

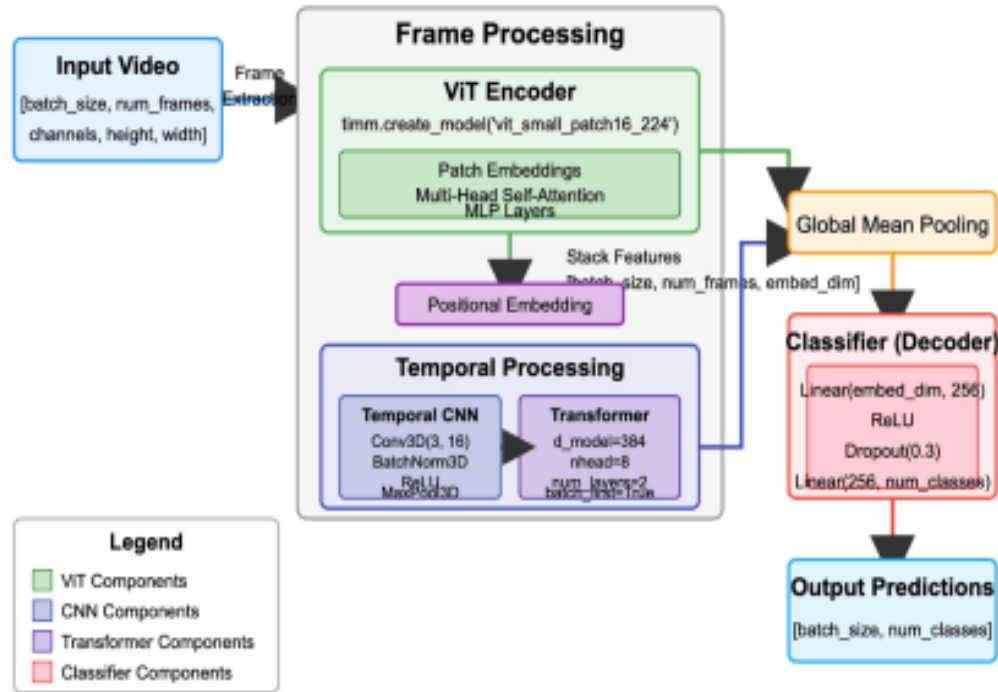


Fig. 2. ViTVideoClassifier Architecture

The frame-wise processing is handled by a ViT encoder, specifically instantiated using `timm.create_model('vit_small_patch16_224')` [14, 16]. Within the ViT, each frame is separated into patches that are embedded and go through several levels of feedforward MLPs and multi-head self-attention. These embeddings encode spatial features and result in frame-wise feature vectors of shape [batch_size, num_frames, embed_dim]. To incorporate the temporal order of frames, the positional embeddings are included in the stacked features. This maintained the awareness of the sequence in which frames appear [13, 14].

Following this, the model enters the temporal processing stage. Here, a Temporal CNN block with 3D convolutional layers captures short-term spatiotemporal dependencies across adjacent frames [4, 6, 8]. The output is then passed into a Transformer block configured with `d_model=384`, `nhead=8`, and `num_layers=2`, which further models long-range temporal dependencies across the video [13, 15]. This combination enables the model to efficiently comprehend global as well as local temporal patterns.

Once temporal processing is complete, a global mean pooling layer is applied across the temporal dimension to condense the information into a single representation vector per video. This pooled feature vector is then passed into a classification head, or decoder, comprising a final linear layer that maps to the number of target classes, a fully linked layer with ReLU activation, and dropout for regularization. A prediction tensor of shape [batch_size, num_classes] is the end product, and it represents the classification scores for every video in the batch.

This architecture effectively combines spatial understanding from Vision Transformers with temporal modeling through CNNs and Transformers, making it well-suited for tasks that require both frame-level detail and linear

reasoning, like deepfake detection in videos or action detection [2, 5, 9, 10, 14].

4.1 Evaluation Measures

We used several evaluation metrics, such as accuracy in classification, Area Under the Curve value, Receiver Operating Characteristic curve and F1-score to assess our model's performance [1, 3, 8]. These measurements offer a thorough insight to differentiate between authentic and edited videos.

Accuracy:

The ratio of correctly classified with the total number of evaluated instances is known as accuracy. It is given as:

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

where, TP - True Positives, FP - False Positives, TN - True Negatives and FN - False Negatives [6, 7].

Precision:

The ratio of correctly predicted positive instances with all predicted positive instances is known as precision. It is also called as positive predictive instance [5, 14]. It is given as:

$$\text{Precision} = TP / (TP + FP)$$

Recall:

Recall is the parameter that calculates classifier's ability to find positive instances. It is also called as sensitivity or true positive rate [3, 14]. It is given as:

$$\text{Recall} = TP / (TP + FN)$$

F1-Score:

The F1-Score is calculated as the mean of Recall and Precision. It especially useful for balancing the class imbalance [7, 8]. It is computed as:

$$\text{F1-Score} = (2 \times \text{Recall} \times \text{Precision}) / (\text{Recall} + \text{Precision})$$

ROC Curve:

The ROC Curve represents binary classification model in graphical format. It shows diagnostic capability of the model with varying discrimination thresholds. The plot of True Positive Rate is drawn with respect to False Positive Rate (FPR) [13, 9]. Here, FPR is given as

$$\text{FPR} = FP / (FP + TN)$$

Area Under the ROC Curve (AUC):

The AUC measures ability of the model to categorize positive and negative classes. Higher the AUC, better is the model performance [14, 10].

Confusion Matrix:

Confusion matrix is 2×2 matrix that provides detailed insight of a classification model. It compares actual and predicted labels and is structured as given [6, 12]:

	Positive (Predicted)	Negative (Predicted)
Positive (Actual)	TP	FN
Negative (Actual)	FP	TN

5. Experimental Outcomes

The decrease in validation accuracy is observed at specific points and specific points, suggesting effective learning

on the training dataset. The model is likely to be overfit after 10-15 epochs because after certain point, validation loss is increased [3, 14].

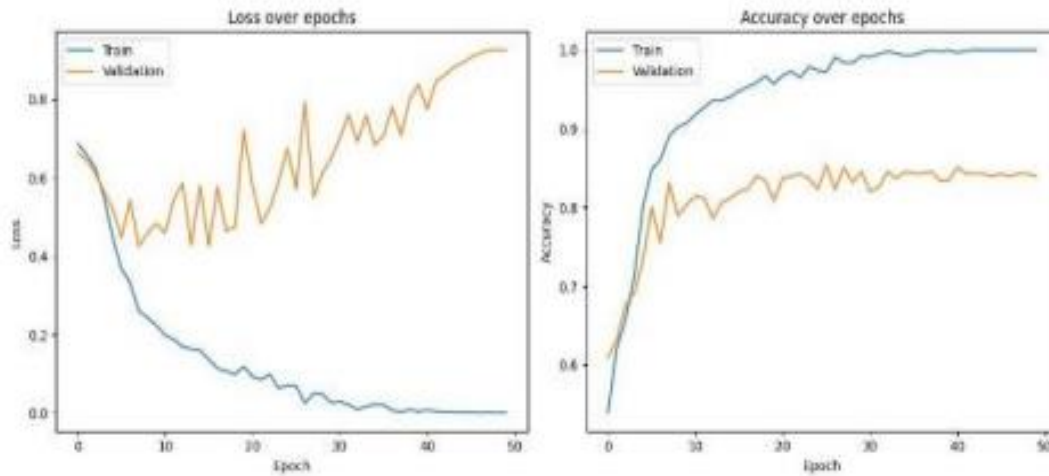


Fig. 3. Loss and Accuracy over epochs

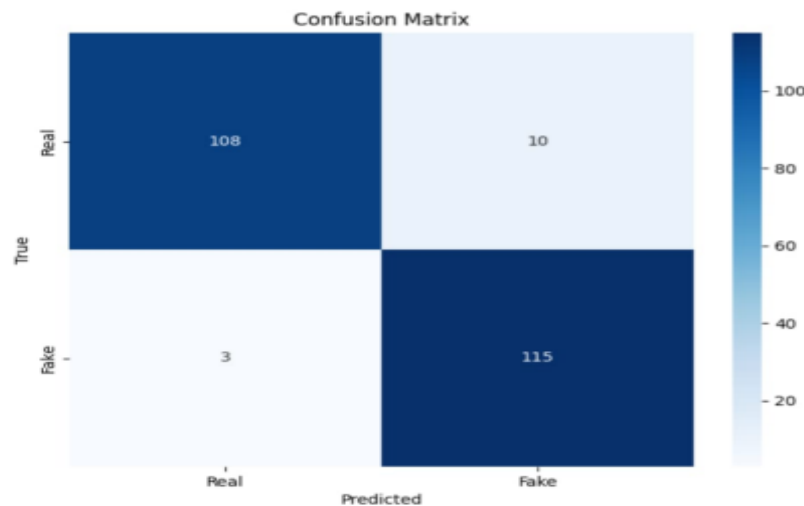


Fig. 4. Confusion Matrix

In Fig. 4, confusion matrix shows that the model accurately recognized 108 genuine videos and 115 fabricated videos, but reveals that 10 genuine videos and 3 fabricated videos are actually counterfeit. The model is very effective as it distinguishes genuine and manipulated content with high accuracy [4, 7].

5.1 Result Analysis

Figure 5 illustrates the comparison of the performances of various ML models with respect to accuracy, recall and precision. The models being compared by a hybrid model (CNN + vision transformer), densenet, LSTM, and SVM. Under these conditions, the hybrid model demonstrates the highest overall performance, achieving a recall rate of approximately 97% and maintaining strong accuracy. The model has a 94 percent accuracy and keep up to an estimated accuracy of 92% [14, 15, 16]. Densenet maintains its consistency and is slightly lower, but is higher on all three

metrics, particularly when recalled at around 95% [3, 8]. The svm model exhibits moderate accuracy in terms of precision and accuracy, with a close approximation of 90%. It has the ability to remember even the smallest details [6, 8]. In comparison, the LSTM model performs lowest for all three metrics, achieving accuracy of approximately 88%, 86%, and 84%. This comparison demonstrates the outstanding performance of hybrid models in classification tasks. This highlights the effectiveness of integrating folding techniques with a transformer-based architecture [14, 15].

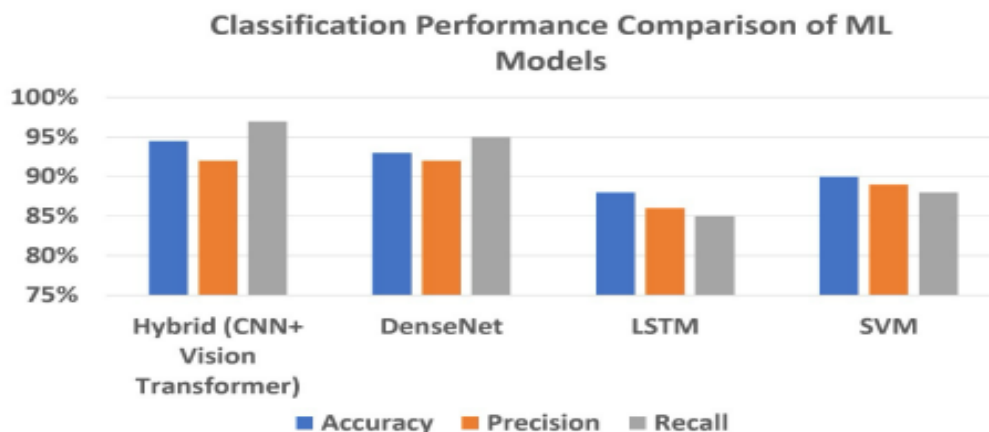


Fig. 5. Comparison of ML Models with respect to accuracy, precision and recall

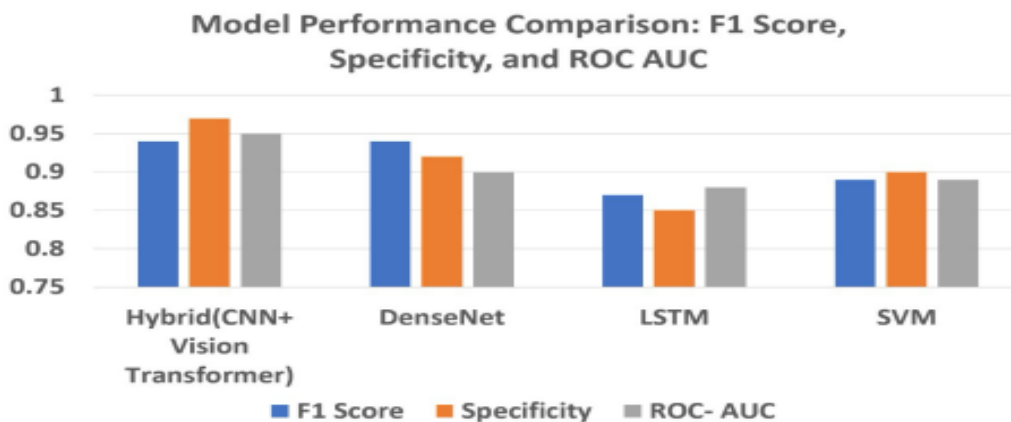


Fig. 6. Model Performance Comparison: F1 Score, Specificity, and ROC AUC

The bar graph showcases a comparative analysis of four machine learning models—hybrid (CNN + vision transformer). The researchers evaluated various machine learning models, including transformer, densenet, LSTM, and SVM, based on three important performance metrics: F1 score and accuracy, Specificity, and ROC AUC. The model's capability to manage datasets with unequal class distributions is remarkable. Among the four models, the hybrid model continuously surpasses the others, attaining the highest values in all three metrics. With a f1 score of 94%, specificity of 97%, and ROC AUC of 95%, it demonstrates exceptional performance [14, 15, 16]. The skill of finding the right balance between being precise and recall, correctly identifying negative cases, and distinguishing them from positive ones.

DenseNet also shows competitive performance, closely trailing the Hybrid model. It maintains a similar F1 Score of around 94%, a Specificity of approximately 92%, and a ROC AUC of about 90%. These results indicate that

Densenet is a robust deep learning model, highly effective in image classification tasks, because of its design, which promotes efficient feature reuse and gradient flow. However, its slightly lower ROC AUC suggests that its ability to separate positive and negative classes is marginally weaker than that of the Hybrid model [14].

In contrast, the LSTM model performs the weakest among all four models. It records the lowest F1 Score (around 87%) and Specificity (approximately 85%), indicating limited effectiveness in correctly identifying negative samples and achieving balanced classification performance. Its ROC AUC score of about 88% is slightly better but still lags behind the CNN-based models [3, 7]. The SVM model, while not as powerful as the deep learning models, shows moderate and consistent results across all metrics, with values close to 89–90%. This makes it a reasonable choice for scenarios with limited computational resources or simpler classification tasks [6, 8].

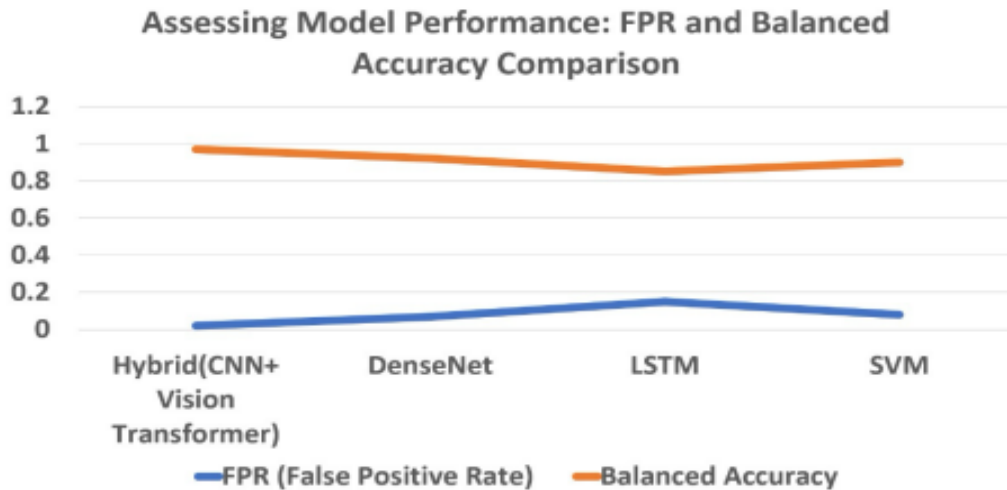


Fig. 7. Assessing Model Performance: FPR and Balanced Accuracy Comparison

The chart compares the performance of four machine learning models: Hybrid (CNN + Vision Transformer), DenseNet, LSTM, and SVM—based on False Positive Rate (FPR) and Balanced Accuracy. The Hybrid model clearly outperforms the rest, exhibiting the lowest FPR and the highest Balanced Accuracy, indicating excellent reliability in correctly classifying both positive and negative samples with minimal false positives [14, 15]. DenseNet also performs well, though slightly behind the Hybrid model in both metrics. In contrast, LSTM shows the poorest performance with the highest FPR and lowest Balanced Accuracy, suggesting a tendency to generate more false positives and imbalanced classification [3]. SVM performs moderately well, surpassing LSTM but falling short of the CNN-based models [6, 8].

5.2 Open Issues and Future Scope:

Despite the progress made in deepfake detection, there are still several unresolved issues that pose challenges to the field:

- Many deepfake detection models excel in performing well on specific datasets, such as faceforensics++ or celeb-df, but struggle to generalize to unseen data or real-world scenarios. The process of improving generalization across various datasets remains a significant challenge [3, 4].
- Real-time detection: the existing methods may not be quick enough for real-time deepfake detection, particularly in high-resolution videos or large-scale applications, which restricts their practical use in scenarios such as live video streams or social media monitoring [4, 10].
- Adversarial robustness: deepfake detection models can be susceptible to adversarial attacks, where minor alterations to deepfake images or videos deceive detection systems. Enhancing the resilience of models against such attacks is crucial for ensuring accurate and dependable detection [1, 3, 5].

- As the quality of deepfake videos improves, it becomes increasingly challenging to detect them. The identification of deepfakes with excellent quality and minimal visible distortions presents a significant challenge for existing detection models [10, 12].
- Scarcity of Fine-Grained Detection Methods: While some methods, like multi-attentional networks, aim to focus on fine-grained details, the majority of detection models still rely on binary classification. Developing more nuanced approaches that can capture subtle, localized features is an ongoing issue [4, 14].
- Dataset Bias: Limited datasets are used for most of the current detection models for training. This may result into biases in terms of ethnicity, gender, or lighting conditions. These biases can reduce detection accuracy in diverse real-world applications [1, 5, 9].
- Privacy and Ethical Concerns: There is a need for guidelines on how detection models interact with privacy laws and the ethical implications of detecting/manipulating media. Ensuring the responsible use of these technologies is an open issue for both developers and policymakers [10, 11, 12].

Addressing these open issues will be key to improving the accuracy, scalability, and robustness of DeepFake detection technologies in the future.

6. Conclusion

In conclusion, a hybrid deep learning model used for AI-generated video authenticity verification addresses a critical challenge in the digital age: the growing threat of misinformation and the spread of fake media. By combining Convolutional Neural Networks with Vision Transformers, the system capitalizes local feature extraction capabilities. It also enhances global context awareness that resulted robust and accurate detection framework. The use of the Leaky Rectified Linear Unit as the activation function further enhances model performance by improving gradient flow and training efficiency.

Empirical results strongly support the superiority of the hybrid approach. The model achieved the highest Recall at approximately 97%, with Accuracy and Precision reaching 94.5% and 92%, respectively—outperforming alternatives such as DenseNet, LSTM, and SVM. Furthermore, the hybrid model exhibited leading results across other critical evaluation metrics, including an F1 Score of 94%, Specificity of 97%, and ROC AUC of 95%, demonstrating its effectiveness in handling imbalanced data, maintaining balanced classification performance, and distinguishing between authentic and manipulated content.

In comparison, DenseNet followed closely with a Recall of 95%, F1 Score of 94%, Specificity of 92%, and ROC AUC of 90% [3], while SVM and LSTM showed moderate to weaker performance. LSTM, in particular, recorded the lowest scores—Accuracy (88%), Precision (86%), Recall (84%), and Specificity (85%)—highlighting the limitations of sequence-based models in spatial-temporal video classification tasks.

This paper demonstrates the potential of hybrid models in effectively combating the dissemination of manipulated media, contributing to the restoration of trust in digital content. The integration of CNNs and ViTs provides a scalable solution that can adapt to a wide range of videos for media, law enforcement, and cybersecurity. Future developments in this area may include refining the model to handle increasingly sophisticated video manipulations and enhancing its real-time processing capabilities, further fortifying its role in the fight against digital misinformation.

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